

Mohammad Ninad Mahmud Nobo

Bangladesh University of Engineering and Technology (BUET) | CSE

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SUMMARY

CSE undergraduate at BUET focused on building intelligent systems at the intersection of large language models, software automation, and applied machine learning. Hands-on experience designing end-to-end AI-driven systems across mobile and backend platforms, with strong emphasis on reliability, scalability, and real-world deployment. Actively engaged in research on LLM-based software testing and medical AI, including dataset creation, evaluation pipelines, and trust-aware decision systems.

EDUCATION

Bangladesh University of Engineering and Technology (BUET)

2022 – Present

Bachelor of Science in Computer Science and Engineering (CSE)

CGPA: 3.61 / 4.00

Relevant Coursework: Data Structures and Algorithms, Digital Logic Design, Database Systems, Operating Systems, Computer Architecture, Computer Networks, Computer Graphics, Algorithm Engineering, Artificial Intelligence, Machine Learning

RESEARCH INTERESTS

- Large Language Models and LLM Systems
- Software Testing Automation
- Applied Machine Learning
- Medical AI and Multimodal Learning

RESEARCH EXPERIENCE

- **Undergraduate Thesis: Web Testing Using Large Language Models** [GitHub Link]

Python • LLMs • Web Automation • Prompt Engineering

2025 – Present

- Designing LLM-based pipelines for automated web test generation using prompt-driven and programmatic approaches.
- Creating and curating a dataset of functional descriptions and corresponding test cases for supervised evaluation.
- Investigating coverage improvement and robustness of generated tests.

- **MedCAR: Conflict-Aware Medical Reasoning System (Under Review)** [GitHub Link]

Python • Deep Learning • Medical AI

2026 – Present

- Built a multi-model system for chest X-ray analysis integrating heterogeneous AI tools.
- Developed a conflict resolution pipeline to reconcile contradictory outputs using semantic reasoning and confidence calibration.
- Designed trust-aware decision framework with uncertainty-based abstention for reliable predictions.

PROJECTS

- **MindTrace: AI-Powered Dementia Care Application** [GitHub Link | Feature Demo | Infrastructure Demo]

Kotlin • Spring Boot • Spring AI • PostgreSQL • Redis • Firebase • Docker • Azure

2025

- Engineered an AI-driven mobile-backend system for dementia care, integrating Android frontend with a scalable Spring Boot architecture supporting real-time caregiver workflows.
- Designed accessibility-first UI/UX for cognitively impaired users, emphasizing clarity, reduced interaction complexity, and usability.

- **Gemma VetCare: AI-Assisted Farming Support System** [GitHub Link | Feature Demo]

Kotlin • Android Studio • Spring Boot • Spring AI • MongoDB

2025

- Built an AI-assisted Android application for veterinary decision support, integrating backend services for real-time recommendations.
- Designed RESTful APIs optimized for low-connectivity environments through efficient request handling and data flow design.

- **Compiler Construction (Flex, Bison, 8086 Codegen)** [GitHub Link]

C++17, Flex, Bison/Yacc • Linux

2024

- Designed and implemented a full compiler pipeline for a C-like language, including lexical analysis, parsing, semantic analysis, and intermediate representation.
- Generated optimized 8086 assembly code with symbol table management and basic optimization techniques.

- **Computer Graphics Pipeline Implementation** [GitHub Link]

C++17 • OpenGL/GLUT • Rasterization • Ray Tracing

2025

- Implemented a full graphics pipeline including geometric transformations, clipping, rasterization with Z-buffering, and ray tracing.
- Developed OpenGL-based rendering demos with camera systems supporting 6-DOF control and real-time scene interaction.

TECHNICAL SKILLS

Languages: C/C++, Python, Java, Kotlin, JavaScript, SQL

Backend: Spring Boot, Spring AI, Node.js, Express.js, REST APIs

AI/ML: PyTorch, TensorFlow, Scikit-learn, LLM Applications

Databases: PostgreSQL, MongoDB, Redis

Tools: Linux, Git, Docker, Firebase

ADDITIONAL INFORMATION

Languages: Bangla (Native), English (Fluent), Hindi & Urdu (Conversational), Arabic (Reading)

Programming Profiles: NeetCode: mninadmno LeetCode: mninadmno